

Lecture 8 Tutorial — Non-parametric Learning

Q1. Non-parametric Modeling (Fill in the blanks)

Complete the statements.

Non-parametric models can be used with arbitrary distributions and without assuming that the forms of the underlying _____ are known.

Non-parametric models are suitable for _____ distributions, which are common in practice.

Q2. Pros and Cons (True / False)

Write T (True) or F (False).

1. Non-parametric methods require no distributional assumptions ahead of time.
2. With enough samples, non-parametric density estimation can converge to an arbitrarily complicated target density.
3. The number of samples needed typically grows linearly with feature dimension.
4. Non-parametric methods can be sensitive to the choice of window size.

Q3. kNN Classification Rule (Multiple Choice)

In kNN classification, the predicted label of a new point is determined by:

- A. the label of the farthest neighbour
- B. the average of neighbour labels
- C. majority vote among the nearest neighbours' labels
- D. maximizing the likelihood of a Gaussian model

Q4. kNN Window / Neighbourhood Interpretation (Short Answer)

The lecture describes a cell centred at _____ that grows until it captures _____ samples.

1. What are those samples called?
2. If the window is too small, what problem may occur?
3. If the window is too large, what problem may occur?

(Answer each in one short phrase.)

Q5. kNN Computation

Use the training set below (from the lecture), where each row is

(Outlook, Temperature, Humidity, Windy, Play)

#	Outlook	Temperature	Humidity	Windy	Play
1	sunny	85	85	1.0	maybe
2	sunny	80	90	3.0	no
3	overcast	83	86	0.8	yes
4	rainy	70	96	0.2	maybe
5	rainy	68	80	0.1	yes
6	rainy	65	70	2.8	no
7	overcast	64	65	2.6	yes
8	sunny	72	95	0.3	no
9	sunny	69	70	0.5	yes
10	rainy	75	80	0.4	maybe
11	sunny	75	70	2.2	yes
12	overcast	72	90	2.4	maybe
13	overcast	81	75	0.0	yes
14	rainy	71	91	2.9	no

We want to predict Play for the case (sunny, 74, 82) as in the lecture prompt.

Assumption (to make the exercise well-defined): use Euclidean distance on (Temperature, Humidity) only, and use $k = 3$.

1. Compute the distances from (74, 82) to each training point (Temperature, Humidity).
2. Identify the 3 nearest neighbours (report their row numbers).
3. Predict the label by majority vote.

Q6. Sensitivity to k (Structured Decision)

Using your neighbour ranking from Q5, answer:

Tie-breaking rule: if there is a tie in votes, choose the class of the closest neighbour.

1. Predicted label for $k = 3$: _____
2. Predicted label for $k = 5$: _____
3. Predicted label for $k = 7$: _____

Q7. Decision Tree Stopping Conditions (Multiple Choice)

A node can be treated as a leaf node (stop splitting) based on lecture-listed checks such as:

- contains no samples,
- samples belong to a universal class,
- a common attribute is shared,
- whether attributes are yet to be further analysed.

Which option best describes how the leaf decision is made?

- A. minimize Euclidean distance
- B. majority vote (mode) of the labels in the node
- C. maximize cluster similarity
- D. pick the attribute with smallest variance

Q8. Entropy (Fill in the blanks)

Complete the two formulas (log base 2):

$$\text{entropy}(D) = - \sum_{k=1}^{|Y|} p_k \log_2 p_k$$

$$\text{Gain}(D, x_j) = \text{entropy}(D) - \sum_{v=1}^V \frac{|D_v|}{|D|} \text{entropy}(D_v)$$

Fill in the missing meanings:

1. p_k is the proportion of class k in _____.
2. D_v is the subset of D with attribute value _____ for _____.

Q9. Entropy Calculation (3-class Node)

A node contains 14 samples with class counts:

$$\text{Yes} = 6, \quad \text{No} = 5, \quad \text{Maybe} = 3.$$

1. Compute $\text{entropy}(D)$ (base 2), rounded to 3 decimal places.
2. State which class is the mode at this node.

Q10. Information Gain for a Split (Computation)

Use the same parent node as Q9. Consider splitting on an attribute with three values, producing:

$$D_1 \text{ (size 6) : } \text{Yes} = 2, \text{ No} = 3, \text{ Maybe} = 1$$

$$D_2 \text{ (size 4) : } \text{Yes} = 3, \text{ No} = 0, \text{ Maybe} = 1$$

$$D_3 \text{ (size 4) : } \text{Yes} = 1, \text{ No} = 2, \text{ Maybe} = 1$$

1. Compute $\text{entropy}(D_1)$, $\text{entropy}(D_2)$, $\text{entropy}(D_3)$.
2. Compute the weighted entropy after the split.
3. Compute $\text{Gain}(D, x_j)$, rounded to 3 decimals.

Q11. Gain Ratio (Computation)

The lecture defines the gain ratio as

$$\text{GainRatio}(D, x_j) = \frac{\text{Gain}(D, x_j)}{\text{IV}(a)}$$

where

$$\text{IV}(a) = - \sum_{v=1}^V \frac{|D_v|}{|D|} \log_2 \left(\frac{|D_v|}{|D|} \right).$$

Using the split in Q10 (sizes 6, 4, 4 out of 14):

1. Compute IV (base 2), rounded to 3 decimals.
2. Using your $\text{Gain}(D, x_j)$ from Q10, compute $\text{GainRatio}(D, x_j)$, rounded to 3 decimals.

Q12. Clustering (Agglomerative + k-means)

Part A — Agglomerative (Hierarchical) Clustering

Using the city-block distance matrix (from the lecture example):

$$d(A, B) = 8, \quad d(A, C) = 3, \quad d(A, D) = 5, \quad d(A, E) = 13$$

$$d(B, C) = 6, \quad d(B, D) = 5, \quad d(B, E) = 10$$

$$d(C, D) = 8, \quad d(C, E) = 2$$

$$d(D, E) = 7$$

Assume single linkage (cluster distance = minimum pairwise distance).

1. What is the first merge?
2. What is the second merge?
3. List the clusters after the second merge.

Part B — One Iteration of k-means

Given 7 points (from the lecture example):

$$(1.0, 1.0), (1.5, 2.0), (3.0, 4.0), (5.0, 7.0), (3.5, 5.0), (4.5, 5.0), (3.5, 4.5)$$

Let $k = 2$ and initialise centroids as:

$$c_1^{(0)} = (1.0, 1.0), \quad c_2^{(0)} = (5.0, 7.0).$$

Use Euclidean distance. If a point is equidistant, assign it to the smaller-index cluster.

1. Assign each point to Cluster 1 or Cluster 2.
2. Compute the updated centroids $c_1^{(1)}$ and $c_2^{(1)}$.